DAV LAB 1

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Class: C1 A2

Roll No: 18

In [21]:
```
import pandas as pd
```

CSV File

In [22]:
```
df_csv=pd.read_csv("data/data.csv")
```

In [23]:
```
print("\nHead")
print(df_csv.head())

print("\nTail")
print(df_csv.tail())

print("\nDescribe")
print(df_csv.describe())

print("\nInfo")
print(df_csv.info())
```

```
Head
   stud_id        stud_name                 branch section
0        1         John Doe        Computer Science       A
1        2       Jane Smith  Electrical Engineering       B
2        3  Michael Johnson  Mechanical Engineering       A
3        4      Emily Davis       Civil Engineering       B
4        5    William Brown        Computer Science       A

Tail
    stud_id             stud_name                 branch section
15       16      Charlotte Garcia       Civil Engineering       B
16       17  Christopher Martinez        Computer Science       A
17       18       Amelia Robinson  Electrical Engineering       B
18       19          Andrew Clark  Mechanical Engineering       A
19       20   Elizabeth Rodriguez       Civil Engineering       B

Describe
        stud_id
count  20.00000
mean   10.50000
std     5.91608
min     1.00000
25%     5.75000
50%    10.50000
75%    15.25000
max    20.00000

Info
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   stud_id    20 non-null     int64
 1   stud_name  20 non-null     str
 2   branch     20 non-null     str
 3   section    20 non-null     str
dtypes: int64(1), str(3)
memory usage: 772.0 bytes
None
```

Txt File

In [24]:
```python
with open("data/hello.txt", 'r') as file:
    content= file.read()
    print(content)
```

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r.

In [25]:
```
df_txt=pd.read_csv("data/hello.txt",delimiter="\t")
df_txt.head()
```

Out[25]:

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JSON File

In [26]:
```
df_json = pd.read_json("data/data.json")
```

In [27]:
```
print("\nHead")
print(df_json.head())

print("\nTail")
print(df_json.tail())

print("\nDescribe")
print(df_json.describe())

print("\nInfo")
print(df_json.info())
```

```
Head
         name  age                email
0    John Doe   30    johndoe@gmail.com
1  Jane Smith   25  janesmith@gmail.com

Tail
         name  age                email
0    John Doe   30    johndoe@gmail.com
1  Jane Smith   25  janesmith@gmail.com

Describe
             age
count   2.000000
mean   27.500000
std     3.535534
min    25.000000
25%    26.250000
50%    27.500000
75%    28.750000
max    30.000000

Info
<class 'pandas.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   name    2 non-null      str
 1   age     2 non-null      int64
 2   email   2 non-null      str
dtypes: int64(1), str(2)
memory usage: 180.0 bytes
None
```

XML File

In [35]:
```
df_xml=pd.read_xml("data/data.xml",parser="etree")
```

In [36]:
```
df_xml.head()
df_xml.tail()
df_xml.info()
df_xml.describe()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 4 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   id          6 non-null      int64
 1   name        6 non-null      str
 2   age         6 non-null      int64
 3   department  6 non-null      str
dtypes: int64(2), str(2)
memory usage: 324.0 bytes
```

Out[36]:

	id	age
count	6.000000	6.000000
mean	103.500000	22.333333
std	1.870829	3.386247
min	101.000000	20.000000
25%	102.250000	20.250000
50%	103.500000	21.500000
75%	104.750000	22.000000
max	106.000000	29.000000

In [30]:
```
import xml.etree.ElementTree as ET
```

In [31]:
```
tree=ET.parse("data/data.xml")
root=tree.getroot()
for e in root.iter():
    print(e.tag,e.text)
```

```
students

student

id 101
name Alice Johnson
age 20
department CS
student

id 102
name John Doe
age 20
department AIML
student

id 103
name Jane Doe
age 29
department AIDS
student

id 104
name Saptanshu Wanjari
age 22
department AIML
student

id 105
name Hardik Sharma
age 21
department CS
student

id 106
name Saumya Khobragade
age 22
department CS
```